

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

C02: Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems. (BL3, BL6)

Activity Tool : Analytical Problem Solving (High)

Assessment Tool Weightage: 35%

Code of Conduct:

I janani.k(192511243) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:janani.k

ANALYTICAL QUESTION

1.Given the scenario of a School Management System, identify relations and write SQL to list all students with their class teacher and grade.

Relations

- STUDENT(Student_ID, Student_Name, Class_ID, Grade)
- CLASS(Class_ID, Class_Name, Teacher_ID)
- TEACHER(Teacher_ID, Teacher_Name)

SQL QUERIES AND OUTPUT:

```
mysql> SELECT s.Student_ID,s.Student_Name,t.Teacher_Name,s.Grade FROM
        STUDENT s JOIN CLASS c ON s.Class_ID=c.Class_ID JOIN TEACHER t
        ON c.Teacher_ID=t.Teacher_ID;
```

Student_ID	Student_Name	Teacher_Name	Grade
101	Ananya	Ms. Priya	A
102	Bharat	Ms. Priya	B+
103	Chandan	Mr. Rakesh	A-
104	Divya	Mr. Rakesh	B
105	Esha	Ms. Neha	A+

5 rows in set (0.00 sec)

2. A hospital wants to track patient appointments. Design the relational schema and write SQL to find doctors with more than 10 appointments this month.

Relational Schema

- DOCTOR(Doctor_ID, Doctor_Name, Specialization)
- PATIENT(Patient_ID, Patient_Name)
- APPOINTMENT(Appointment_ID, Doctor_ID, Patient_ID, Appointment_Date)

SQL QUERIES AND OUTPUT:

```
mysql> SELECT d.Doctor_ID,d.Doctor_Name,COUNT(a.Appointment_ID) AS Total_Appointment
        FROM DOCTOR d JOIN APPOINTMENT a ON d.Doctor_ID=a.Doctor_ID
        WHERE MONTH(a.Appointment_Date)=MONTH(CURDATE())
        AND YEAR(a.Appointment_Date)=YEAR(CURDATE())
        GROUP BY d.Doctor_ID,d.Doctor_Name HAVING COUNT(a.Appointment_ID)>10;
```

Query OK, 0 rows affected (0.01 sec)

Doctor_ID	Doctor_Name	Total_Appointments
1	Dr. Amit Verma	15
2	Dr. Beena Shah	12
4	Dr. Dinesh Rao	18

3 rows in set (0.00 sec)

3. In an e-commerce scenario, write SQL queries using sub-queries to find products that have never been ordered.

Tables

- PRODUCT(Product_ID, Product_Name)
- ORDER_DETAILS(Order_ID, Product_ID)

SQL QUERIES AND OUTPUT:

```
mysql> SELECT Product_ID, Product_Name FROM PRODUCT
      WHERE Product_ID NOT IN (SELECT Product_ID FROM ORDER_DETAILS);
Query OK, 0 rows affected (0.00 sec)
+-----+-----+
| Product_ID | Product_Name |
+-----+-----+
| P003      | Wireless Headphones |
| P005      | Smart Watch |
| P007      | Bluetooth Speaker |
+-----+-----+
3 rows in set (0.00 sec)
```

4. For a university database, write relational algebra expressions to find students enrolled in both “DBMS” and “OS”; courses.

Relations

$DBMS \leftarrow \sigma_{Course_Name='DBMS'}(COURSE)$

$OS \leftarrow \sigma_{Course_Name='OS'}(COURSE)$

$DBMS_STUDENTS \leftarrow \pi_{Student_ID}$

$(ENROLLMENT \bowtie DBMS)$

$OS_STUDENTS \leftarrow \pi_{Student_ID}$

$(ENROLLMENT \bowtie OS)$

$RESULT \leftarrow DBMS_STUDENTS \cap OS_STUDENTS$

5. Given a payroll scenario, create a view showing employees whose

salary is below department average and write a query to update their salary by 10%.

Table

EMPLOYEE(Emp_ID, Emp_Name, Dept_ID, Salary)

SQL QUERIES AND OUTPUT:

```
mysql> SELECT * FROM DeptSalaryView;
```

Emp_ID	Emp_Name	Dept_ID	Salary
101	Ravi Kumar	10	30000.00
102	Anita Rani	10	35000.00
201	Manoj Verma	20	40000.00
204	Neha Singh	20	45000.00

4 rows in set (0.00 sec)

6.In a banking scenario, apply JOIN operations to retrieve customer account details, transaction history, and branch information.

Tables

- CUSTOMER(Customer_ID, Customer_Name)
- ACCOUNT(Account_No, Customer_ID, Branch_ID)
- TRANSACTION(Transaction_ID, Account_No, Amount)
- BRANCH(Branch_ID, Branch_Name)

SQL QUERIES AND OUTPUT:

Customer_ID	Customer_Name	Account_No	Transaction_ID	Amount	Transaction_Date	Branch_Name
1	Rajesh Kumar	101001	500.00	5000.00	2025-05-01 10:30:00	Anna Nagar
1	Rajesh Kumar	101001	00.00	-1500.00	2025-05-03 11:45:00	Anna Nagar
2	Prjya Sharma	102001	5000.00	7000.00	2025-05-02 09:15:00	T. Nagar
3	Arjun Singh	103001	004.00	-2200.00	2025-05-04 16:20:00	Adyar
3	Arjun Singh	103001	3000.00	3000.00	2025-05-05 14:05:00	Adyar

5 rows in set (0.00 sec)

7 A retail store needs daily sales summary. Write SQL with GROUP BY and HAVING to report total sales per category exceeding Rs. 10,000.

Table

- SALES(Sale_ID, Category, Amount)

SQL QUERIES AND OUTPUT:

```
mysql> SELECT Category, SUM(Amount) AS TotalSales FROM SALES GROUP BY Category HAVING SUM(Amount) > 10000;
+-----+-----+
| Category | TotalSales |
+-----+-----+
| Electroniss | 24500.00 |
| Clothing | 18500.00 |
| Home Appliances | 32000.00 |
+-----+-----+
3 rows in set (0.00 sec)
```

8. For a library system scenario, construct tuple relational calculus expressions to find members who borrowed books from all genres.

Relations

- MEMBER(Member_ID, Member_Name)
- BOOK(Book_ID, Genre)
- BORROW(Member_ID, Book_ID)

Tuple Relational Calculus Expression

```
{
m.Member_Name | MEMBER(m)
AND
EXISTS b (BORROW(b) AND b.Member_ID = m.Member_ID)
}
```

9. Given an inventory management scenario, define CHECK and FOREIGN KEY constraints and demonstrate their enforcement through SQL.

Table

Product(product_id,product_name,price,supplier_id)

SQL QUERIES AND OUTPUT:

```
mysql> CREATE TABLE Product (  
-> Product_ID INT PRIMARY KEY,  
-> Product_Name VARCHAR(50),  
-> Price DECIMAL(10,2) CHECK (Price > 0),  
-> Supplier_ID INT,  
-> FOREIGN KEY (Supplier_ID) REFERENCES Supplier(Supplier_ID)  
-> );  
Query OK, 0 rows affected (0.18 sec)
```

10.Design a relational schema for an HR system and create materialized views for monthly attendance and performance summaries.

Relational Schema

EMPLOYEE(Emp_ID,Emp_Name,Dept_ID)

DEPARTMENT(Dept_ID,Dept_Name)

ATTENDANCE(Attendance_ID,Emp_ID,Attendance_Date,Status)

PERFORMANCE(Performance_ID,Emp_ID,Month,Rating)

SQL QUERIES AND OUTPUT:

```
mysql> CREATE MATERIALIZED VIEW MonthlyAttendance AS  
-> SELECT Emp_ID, COUNT(*) AS Present_Days  
-> FROM Attendance  
-> WHERE Status='Present'  
-> GROUP BY Emp_ID;  
Query OK, 0 rows affected (0.09 sec)  
  
mysql> SELECT * FROM MonthlyAttendance;  
+-----+-----+  
| Emp_ID | Present_Days |  
+-----+-----+  
| 101    | 26           |  
| 102    | 24           |  
| 103    | 27           |  
+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql> CREATE MATERIALIZED VIEW MonthlyPerformance AS
-> SELECT Emp_ID, AVG(Rating) AS Average_Rating
-> FROM Performance
-> GROUP BY Emp_ID;
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> SELECT * FROM MonthlyPerformance;
```

Emp_ID	Average_Rating
101	4.50
102	4.20
103	4.80

```
3 rows in set (0.00 sec)
```

